Lab Assignment 3.3

Task 1 :

Prompt1 : Write a python program for an electricity billing system that reads previous units and current units and type of customer it should calculate units consumed. It should implements logic directly in main program and the energy charges depend on number of units consumed and customer type .

Code1:

```
def calculate_bill(previous_units: int, current_units: int, customer_type: str) ->
float:
    units_consumed = current_units - previous_units
    if units_consumed < 0:
        raise ValueError("Current units must be greater than or equal to previous
units.")


    if customer_type.lower() == "residential":
        if units_consumed <= 100:
            rate = 0.5
        elif units_consumed <= 300:
            rate = 0.75
        else:
            rate = 1.0
    elif customer_type.lower() == "commercial":
        if units_consumed <= 100:
            rate = 0.75
        elif units_consumed <= 300:
            rate = 1.0
```

```
        else:
            rate = 1.5
    else:
        raise ValueError("Invalid customer type. Must be 'residential' or
'commercial'.")


    return units_consumed * rate
if __name__ == "__main__":
    try:
        previous_units = int(input("Enter previous meter reading: ").strip())
        current_units = int(input("Enter current meter reading: ").strip())
        customer_type = input("Enter customer type (residential/commercial):
").strip()


        bill_amount = calculate_bill(previous_units, current_units, customer_type)
        print(f"Total bill amount: ${bill_amount:.2f}")
    except ValueError as e:
        print(f"Error: {e}")
        raise SystemExit(1)
```

output:

```
Enter previous meter reading: 5000
Enter current meter reading: 6500
Enter customer type (residential/commercial): residential
Total bill amount: $1500.00
```

Task 2:

Prompt2: In above code energy charges depend on number of units consumed and customer type .now it should calculate energy charge on domestic commercial industrial consumers

Code:

```
def calculate_bill(previous_units: int, current_units: int, customer_type: str) -> float:
    units_consumed = current_units - previous_units
    if units_consumed < 0:
        raise ValueError("Current units must be greater than or equal to previous units.")


    if customer_type.lower() == "residential":
        if units_consumed <= 100:
            rate = 0.5
        elif units_consumed <= 300:
            rate = 0.75
        else:
            rate = 1.0
    elif customer_type.lower() == "commercial":
        if units_consumed <= 100:
            rate = 0.75
        elif units_consumed <= 300:
            rate = 1.0
        else:
            rate = 1.5
    elif customer_type.lower() == "industrial":
        if units_consumed <= 100:
            rate = 1.0
        elif units_consumed <= 300:
            rate = 1.5
        else:
```

```
        rate = 2.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or 'industrial'.")


    return units_consumed * rate
if __name__ == "__main__":
    try:
        previous_units = int(input("Enter previous meter reading: ").strip())
        current_units = int(input("Enter current meter reading: ").strip())
        customer_type = input("Enter customer type (residential/commercial/industrial): ").strip()


        bill_amount = calculate_bill(previous_units, current_units, customer_type)
        print(f"Total bill amount: ${bill_amount:.2f}")
    except ValueError as e:
        print(f"Error: {e}")
        raise SystemExit(1)
```

output:

Enter previous meter reading: 5000

Enter current meter reading: 6500

Enter customer type (residential/commercial/industrial): commercial

Total bill amount: $2250.00

Task 3:

Prompt3: In above code billing logic must be reusable for multiple consumers it should use defined functions to calculate energy charges, fixed charges and it return calculated value.

Code :

```
def calculate_energy_charges(units_consumed: int, customer_type: str) -> float:
    if customer_type.lower() == "residential":
        if units_consumed <= 100:
            rate = 0.5
        elif units_consumed <= 300:
            rate = 0.75
        else:
            rate = 1.0
    elif customer_type.lower() == "commercial":
        if units_consumed <= 100:
            rate = 0.75
        elif units_consumed <= 300:
            rate = 1.0
        else:
            rate = 1.5
    elif customer_type.lower() == "industrial":
        if units_consumed <= 100:
            rate = 1.0
        elif units_consumed <= 300:
            rate = 1.5
        else:
            rate = 2.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")


    return units_consumed * rate
def calculate_fixed_charges(customer_type: str) -> float:
```

```
    if customer_type.lower() == "residential":
        return 50.0
    elif customer_type.lower() == "commercial":
        return 100.0
    elif customer_type.lower() == "industrial":
        return 150.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")
def calculate_bill(previous_units: int, current_units: int, customer_type: str) -> float:
    units_consumed = current_units - previous_units
    if units_consumed < 0:
        raise ValueError("Current units must be greater than or equal to previous units.")

    energy_charges = calculate_energy_charges(units_consumed, customer_type)
    fixed_charges = calculate_fixed_charges(customer_type)

    return energy_charges + fixed_charges
if __name__ == "__main__":
    try:
        previous_units = int(input("Enter previous meter reading: ").strip())
        current_units = int(input("Enter current meter reading: ").strip())
        customer_type = input("Enter customer type (residential/commercial/industrial):
").strip()

        bill_amount = calculate_bill(previous_units, current_units, customer_type)
        print(f"Total bill amount: ${bill_amount:.2f}")
    except ValueError as e:
        print(f"Error: {e}")
        raise SystemExit(1)
```

output: Enter previous meter reading: 5000

Enter current meter reading: 6500

Enter customer type (residential/commercial/industrial): residential

Total bill amount: $1550.00

Task 4:

Prompt: # In above code it should include multiple additional charges liked fixedcharge , customercharge , electrictyduty.

Code:

```
def calculate_energy_charges(units_consumed: int, customer_type: str) -> float:
    if customer_type.lower() == "residential":
        if units_consumed <= 100:
            rate = 0.5
        elif units_consumed <= 300:
            rate = 0.75
        else:
            rate = 1.0
    elif customer_type.lower() == "commercial":
        if units_consumed <= 100:
            rate = 0.75
        elif units_consumed <= 300:
            rate = 1.0
        else:
            rate = 1.5
    elif customer_type.lower() == "industrial":
        if units_consumed <= 100:
            rate = 1.0
        elif units_consumed <= 300:
            rate = 1.5
```

```
        else:
            rate = 2.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")


    return units_consumed * rate
def calculate_fixed_charges(customer_type: str) -> float:
    if customer_type.lower() == "residential":
        return 50.0
    elif customer_type.lower() == "commercial":
        return 100.0
    elif customer_type.lower() == "industrial":
        return 150.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")
def calculate_customer_charge(customer_type: str) -> float:
    if customer_type.lower() == "residential":
        return 20.0
    elif customer_type.lower() == "commercial":
        return 40.0
    elif customer_type.lower() == "industrial":
        return 60.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")
def calculate_electricity_duty(bill_amount: float) -> float:
    return bill_amount * 0.05  # 5% electricity duty
def calculate_bill(previous_units: int, current_units: int, customer_type: str) -> float:
```

```
    units_consumed = current_units - previous_units
    if units_consumed < 0:
        raise ValueError("Current units must be greater than or equal to previous units.")

    energy_charges = calculate_energy_charges(units_consumed, customer_type)
    fixed_charges = calculate_fixed_charges(customer_type)
    customer_charge = calculate_customer_charge(customer_type)

    subtotal = energy_charges + fixed_charges + customer_charge
    electricity_duty = calculate_electricity_duty(subtotal)

    return subtotal + electricity_duty
if __name__ == "__main__":
    try:
        previous_units = int(input("Enter previous meter reading: ").strip())
        current_units = int(input("Enter current meter reading: ").strip())
        customer_type = input("Enter customer type (residential/commercial/industrial): ").strip()

        bill_amount = calculate_bill(previous_units, current_units, customer_type)
        print(f"Total bill amount: ${bill_amount:.2f}")
    except ValueError as e:
        print(f"Error: {e}")
        raise SystemExit(1)
```

output: Enter previous meter reading: 5000

Enter current meter reading: 6500

Enter customer type (residential/commercial/industrial): residential

Total bill amount: $1648.50

Task 5:

Prompt: #In above code it should print energy charges , fixed charges , customer charge ,electricity duty ,total bill amount.

Code:

```
def calculate_energy_charges(units_consumed: int, customer_type: str) -> float:
    if customer_type.lower() == "residential":
        if units_consumed <= 100:
            rate = 0.5
        elif units_consumed <= 300:
            rate = 0.75
        else:
            rate = 1.0
    elif customer_type.lower() == "commercial":
        if units_consumed <= 100:
            rate = 0.75
        elif units_consumed <= 300:
            rate = 1.0
        else:
            rate = 1.5
    elif customer_type.lower() == "industrial":
        if units_consumed <= 100:
            rate = 1.0
        elif units_consumed <= 300:
            rate = 1.5
        else:
            rate = 2.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or 'industrial'.")
```

```
    return units_consumed * rate
def calculate_fixed_charges(customer_type: str) -> float:
    if customer_type.lower() == "residential":
        return 50.0
    elif customer_type.lower() == "commercial":
        return 100.0
    elif customer_type.lower() == "industrial":
        return 150.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")
def calculate_customer_charge(customer_type: str) -> float:
    if customer_type.lower() == "residential":
        return 20.0
    elif customer_type.lower() == "commercial":
        return 40.0
    elif customer_type.lower() == "industrial":
        return 60.0
    else:
        raise ValueError("Invalid customer type. Must be 'residential', 'commercial', or
'industrial'.")
def calculate_electricity_duty(bill_amount: float) -> float:
    return bill_amount * 0.05  # 5% electricity duty
def calculate_bill(previous_units: int, current_units: int, customer_type: str) -> float:
    units_consumed = current_units - previous_units
    if units_consumed < 0:
        raise ValueError("Current units must be greater than or equal to previous units.")


    energy_charges = calculate_energy_charges(units_consumed, customer_type)
    fixed_charges = calculate_fixed_charges(customer_type)
```

```
    customer_charge = calculate_customer_charge(customer_type)

    subtotal = energy_charges + fixed_charges + customer_charge
    electricity_duty = calculate_electricity_duty(subtotal)

    total_bill = subtotal + electricity_duty

    return energy_charges, fixed_charges, customer_charge, electricity_duty, total_bill
if __name__ == "__main__":
    try:
        previous_units = int(input("Enter previous meter reading: ").strip())
        current_units = int(input("Enter current meter reading: ").strip())
        customer_type = input("Enter customer type (residential/commercial/industrial): ").strip()

        energy_charges, fixed_charges, customer_charge, electricity_duty, total_bill = calculate_bill(previous_units, current_units, customer_type)

        print(f"Energy Charges: ${energy_charges:.2f}")
        print(f"Fixed Charges: ${fixed_charges:.2f}")
        print(f"Customer Charge: ${customer_charge:.2f}")
        print(f"Electricity Duty: ${electricity_duty:.2f}")
        print(f"Total Bill Amount: ${total_bill:.2f}")
    except ValueError as e:
        print(f"Error: {e}")
        raise SystemExit(1)
```

output:

Enter previous meter reading: 5000

Enter current meter reading: 6000

Enter customer type (residential/commercial/industrial): residential

Energy Charges: $1000.00

Fixed Charges: $50.00

Customer Charge: $20.00

Electricity Duty: $53.50

Total Bill Amount: $1123.50